



UNIT III: DISASTER MANAGEMENT IN INDIA

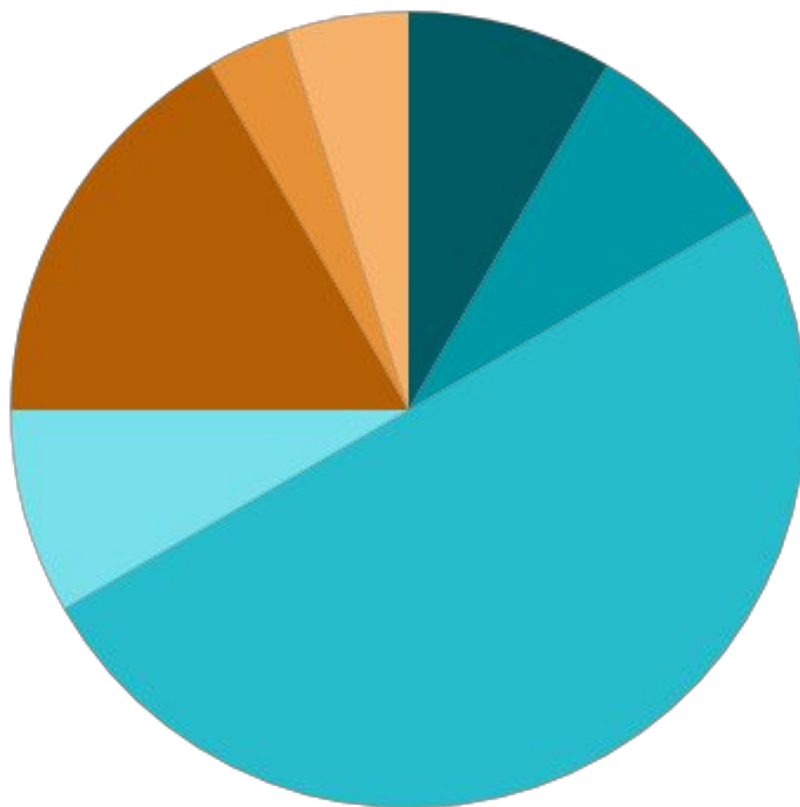


Lecture Outline — Disaster Management in India

- **India's Disaster Profile**
- **Geographical, Geological, Climatic and Socio-Economic Factors**
- **Major Disasters in India and Lessons Learned**
 - Bhuj Earthquake (2001)
 - Uttarakhand Disaster (2013)
 - Kerala Floods (2018)
- **Disaster Management Act, 2005**
- **Institutional Structure of Disaster Management**

Roles of NDMA, SDMA, DDMA and NDRF.
- **Role of Government and NGOs in Disaster Management**

Lecture Structure



05 Minutes

Outline & Objectives



05 Minutes

Recap of Previous Lecture



30 Minutes

Delivery of Actual Content as per Lesson Plan



05 Minutes

Industry Relevance



10 Minutes

Interactive Activities



02 Minutes

Importance of Discipline



03 Minutes

Recording of Attendance

Reference: Office Order No. Provost/092023135, Date: 18th September 2023



Learning Objectives

After completing this lecture, students will be able to:

1. **Explain India's disaster vulnerability** and identify the major natural hazards affecting the country.
2. **Describe the geographic, climatic, and socio-economic factors** that increase disaster risks in India.
3. **Analyze major disaster events in India** and understand the lessons learned from them.
4. **Understand the Disaster Management Act, 2005** and its role in transforming India's disaster management system.
5. **Explain the institutional framework of disaster management** including NDMA, SDMA, DDMA, and NDRF.
6. **Discuss the roles of government agencies, local authorities, and NGOs** in disaster preparedness, response, and recovery.



Recap of Previous Lecture

1. Highlights of previous lecture
2. Interactive Questions

Introduction & Background

India is one of the **most disaster-prone countries in the world** due to its diverse geography, climatic conditions, and high population density.

Several natural hazards frequently affect the country, including:

Earthquakes; Floods; Cyclones; Landslides; Droughts; Heat waves.

India's vulnerability to disasters can be understood through the following statistics:

- **59% of land area lies in earthquake-prone zones**
- **Nearly 40 million hectares are flood-prone**
- **About 68% of land is vulnerable to drought**
- **Over 7,500 km of coastline is exposed to cyclones**
- **Millions of people live in hazard-prone regions**



Introduction & Background

Historically, disaster management in India focused mainly on **relief and rehabilitation after disasters occurred**.

However, after major disasters such as the **Bhuj Earthquake (2001)** and the **Indian Ocean Tsunami (2004)**, India adopted a **proactive disaster management approach** emphasizing:

- Risk reduction
- Preparedness
- Early warning systems
- Community participation.

This shift was formalized through the **Disaster Management Act, 2005**, which established a comprehensive national disaster management framework.

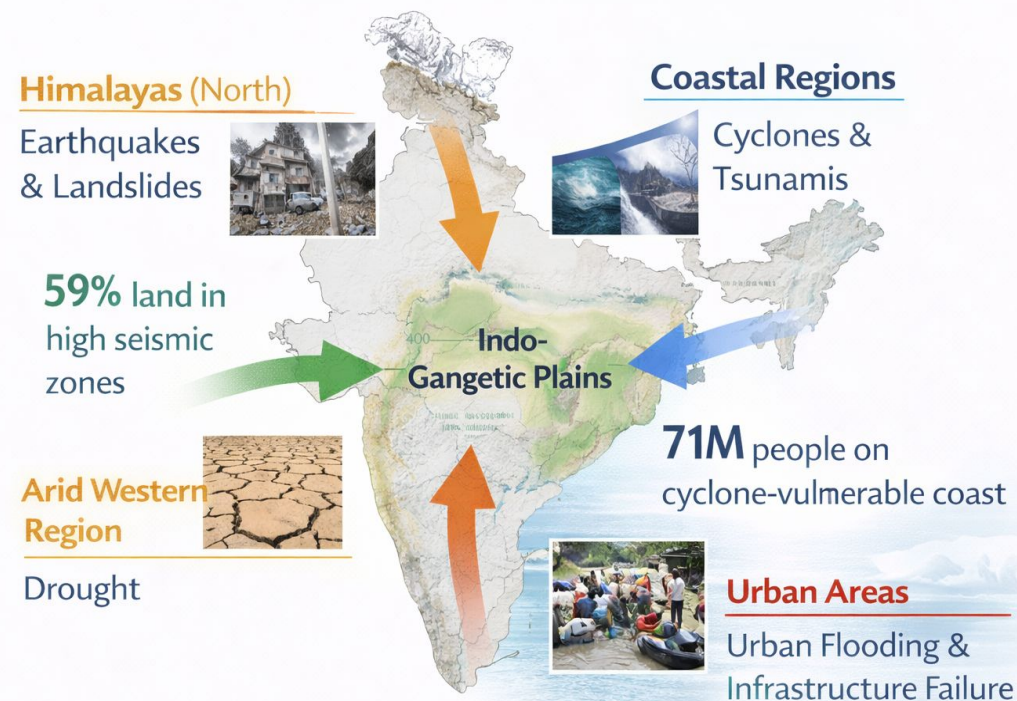
Why India is Highly Disaster-Prone — Five Interlocking Factors

India is among the most disaster-prone countries due to a combination of **geographical, geological, climatic, and socio-economic factors**.

Key national vulnerability indicators:

- **59% of India's land area** lies in **high to very high seismic zones**
- Nearly **40 million hectares** of land is **flood-prone**
- Around **68% of land area** is **arid or semi-arid**, vulnerable to drought
- India has a **7,516 km long coastline** exposed to **cyclones and storm surges**
- Approximately **71 million people** live in **cyclone-vulnerable coastal areas**

India's Multi-Hazard Vulnerabilities



Disasters in India are therefore not caused by a single factor but by **multiple interacting natural and human influences**.

Geographic Factors Increasing Disaster Risk

Himalayan Region

- Young and unstable mountain system
- Prone to **earthquakes, landslides, avalanches**
- Presence of **glacial lakes causing GLOF hazards**

Indo-Gangetic Plains

- Dense population and fertile alluvial plains
- Major rivers such as **Ganga and Brahmaputra**
- Highly vulnerable to **seasonal flooding**

Long Coastline

- India has a **7,516 km coastline**
- Exposed to **cyclones, storm surges, and tsunamis**

Western Arid Region

- Rajasthan and parts of Gujarat experience **chronic drought**
- Water scarcity and desertification are major concern



Geological Causes of Disasters in India

India's geological structure plays a critical role in earthquake occurrence.

Plate Tectonics

- The **Indian Plate** is moving northwards
- It collides with the **Eurasian Plate**
- This collision created the **Himalayan mountain range**

This tectonic movement generates **seismic stress**, making the Himalayan region highly earthquake-prone.

Active Fault Lines

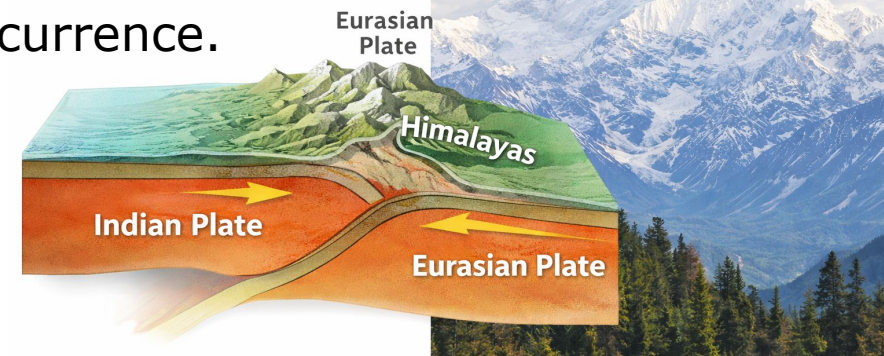
Several fault systems exist across the country, increasing earthquake risk.

Example:

- **Latur Earthquake (1993)** occurred in peninsular India, demonstrating that earthquakes can occur even in stable regions.

Volcanic Activity

India's only active volcano: **Barren Island Volcano** in the Andaman & Nicobar Islands.



Indian Plate colliding with Eurasian Plate → Himalayan seismic zone

Climatic Factors Influencing Disasters

India's climate is dominated by the **monsoon system**, which significantly affects disaster patterns.

Monsoon Dependency

- Nearly **70–80% of annual rainfall occurs between June and September**
- Variability in monsoon leads to extreme events.

Possible outcomes:

- **Excess rainfall → Floods and landslides**
- **Monsoon failure → Drought**

Cyclone Formation

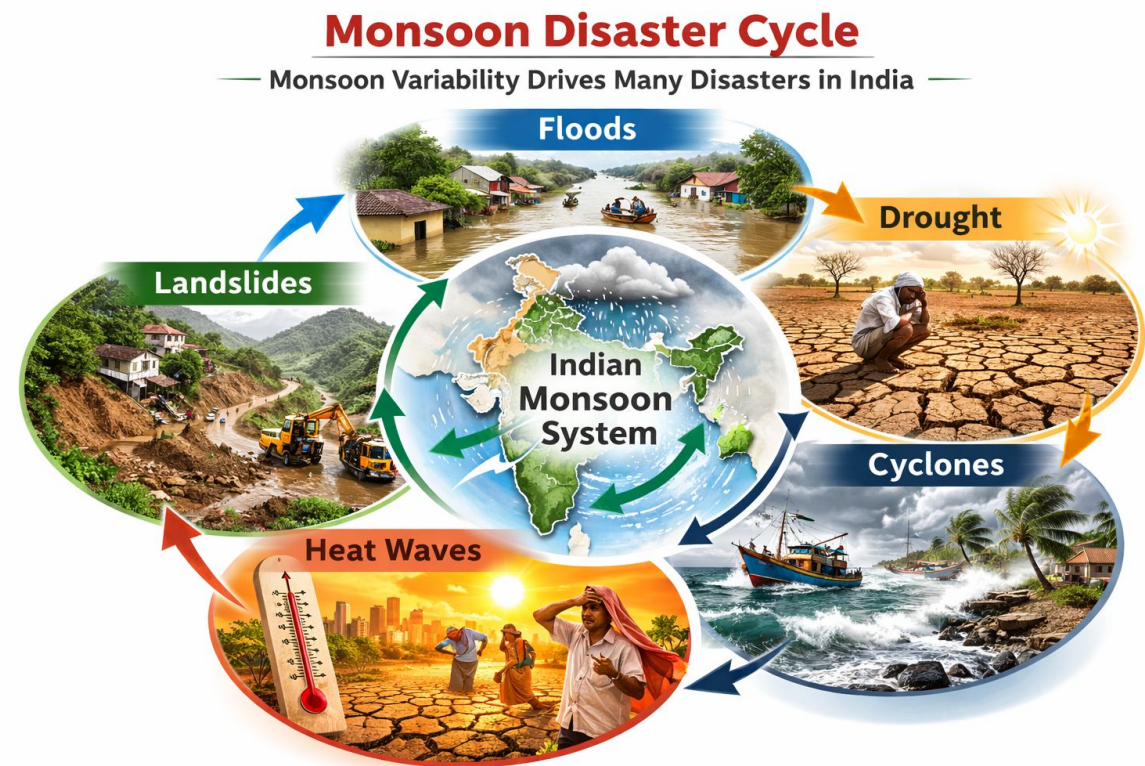
- Bay of Bengal generates **4–5 cyclones annually**
- Arabian Sea generates **1–2 cyclones**

Heat Waves

- Temperatures often exceed **45°C**
- Increasing due to climate change

Cloudbursts

- Sudden extreme rainfall events (>100 mm/hour) - Common in Himalayan regions.



Socio-Economic Factors Increasing Disaster Impact

Human and socio-economic conditions greatly increase the severity of disasters.

Population Density

- India has one of the **highest population densities in the world**
- More people exposed to hazards

Poverty

- Poor households often live in **weak housing structures**
- Limited access to insurance or financial protection

Rapid Urbanization

- Growth of **informal settlements**
- Many located in **floodplains or hazardous areas**

Infrastructure Gaps

- Old buildings without seismic design
- Inadequate drainage systems causing **urban floods**

These factors transform natural hazards into **major disasters affecting millions of people.**



Major Disasters in India — Learning from Experience

India's disaster management system has evolved significantly by learning from **major disaster events**. Each disaster revealed weaknesses in preparedness and response systems and led to important policy and institutional reforms.

Major disasters that transformed disaster management policies include:

- **Bhuj Earthquake (2001)**
 - Led to the **Disaster Management Act 2005**
- **Odisha Super Cyclone (1999)**
 - Improved cyclone preparedness and early warning systems
- **Uttarakhand Floods (2013)**
 - Highlighted risks of unplanned development in fragile ecosystems
- **Kerala Floods (2018)**
 - Emphasized dam management and climate resilience
- **COVID-19 Pandemic (2020)**
 - Demonstrated the need for biological disaster preparedness

These events shifted India's disaster management approach from **reactive relief to proactive risk reduction**.



Bhuj Earthquake (2001) — A Transformative Disaster

The **Bhuj Earthquake** occurred on **26 January 2001** in Gujarat and remains one of the most devastating earthquakes in India.

Key facts:

- **Magnitude:** 7.7 Mw
- **Depth:** ~23 km
- **Deaths:** ~20,000 people
- **Injuries:** ~167,000
- **People affected:** ~6 million
- **Houses destroyed/damaged:** over **1 million**

Major cities such as **Bhuj, Anjar, and Bachau** were almost completely destroyed.

Economic loss was estimated at **₹60,000 crores**.

The disaster exposed serious weaknesses in:

- Building construction practices
- Emergency response systems
- Disaster preparedness planning.



Key Lessons from the Bhuj Earthquake

1. Disaster Management Act (2005)

Established a legal framework for disaster management in India.

2. Creation of NDMA

National Disaster Management Authority headed by the Prime Minister.

3. Formation of NDRF

Specialized disaster response force trained in:

- Urban search and rescue
- Flood rescue
- Chemical and industrial accidents.

4. Enforcement of Building Codes

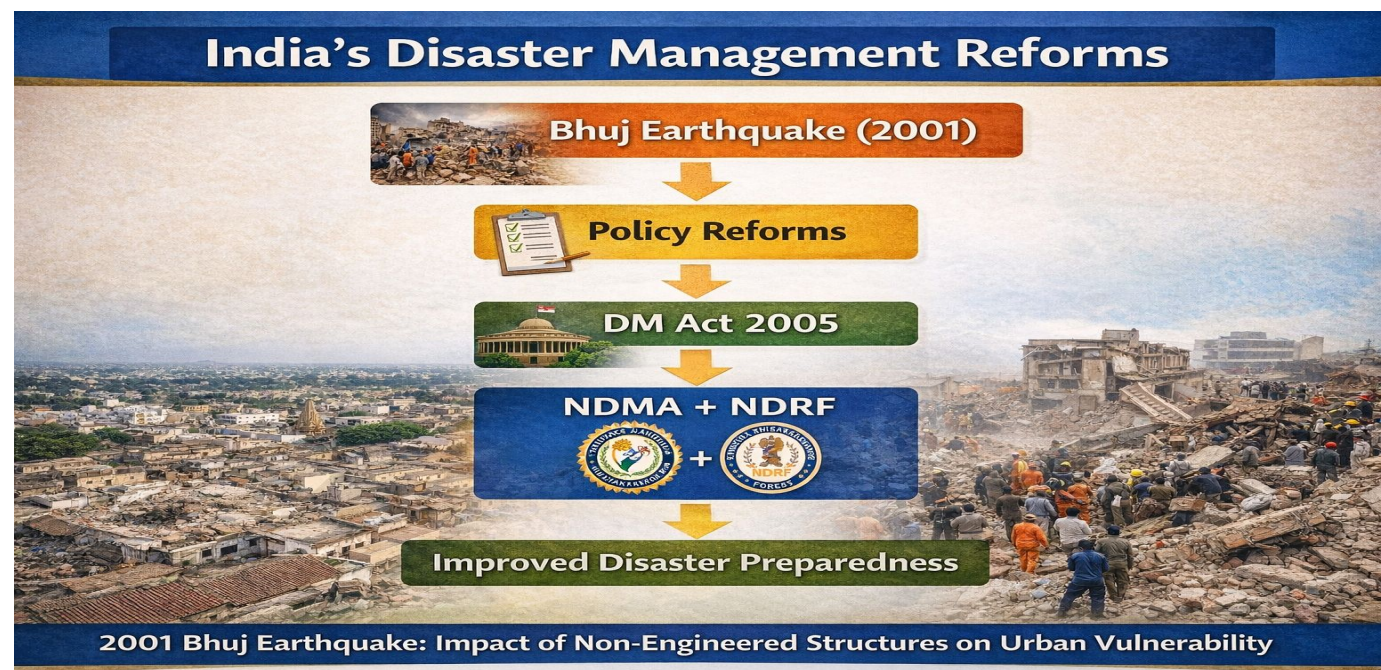
Important codes introduced:

- **IS 1893** — earthquake-resistant design
- **IS 4326** — structural safety guidelines.

5. Community-Based Disaster Management

Local communities trained to participate in disaster preparedness.

These reforms significantly improved India's disaster response capacity.



Uttarakhand Disaster (2013) – Flash Floods & Landslides

The **Uttarakhand disaster of June 2013** was caused by extreme rainfall and glacial lake flooding.

Key facts:

- **Rainfall:** 340 mm in 24 hours
- **Deaths:** over 5,700 people
- **Pilgrims stranded:** ~100,000
- **Roads damaged:** over 1,500 km
- **Bridges destroyed:** more than 200

The disaster severely affected the **Char Dham pilgrimage region**, including Kedarnath.

Major causes included:

- Cloudbursts
- Glacial lake outburst flood (GLOF)
- Landslides
- Unplanned construction along riverbanks.



Lessons from the Uttarakhand Disaster

The disaster highlighted the environmental vulnerability of the Himalayan region.

Key lessons learned:

1. Regulating Development

- Construction along riverbanks must be controlled
- Ecological carrying capacity should be respected.

2. Monitoring Glacial Lakes

- Over **200 glacial lakes** identified as potential threats
- Satellite monitoring introduced.

3. Early Warning Systems

- Installation of **automatic weather stations**
- Improved weather forecasting by IMD.

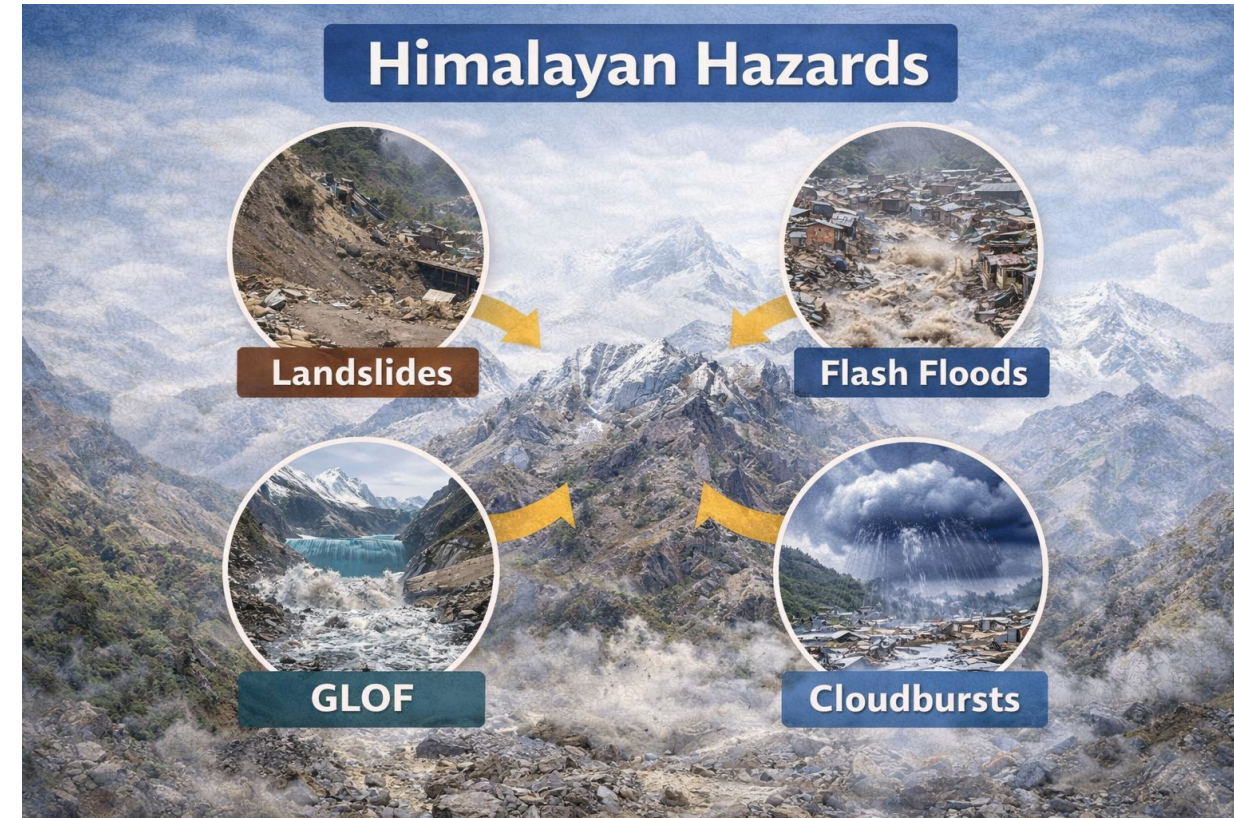
4. Disaster-Resilient Tourism

- Controlled pilgrimage numbers
- Improved evacuation infrastructure.

5. Communication Systems

- Satellite phones and HAM radios introduced as backup communication systems.

These measures aim to reduce future disaster risk in the Himalayan region.



Kerala Floods (2018) – Climate and Infrastructure Challenge

The **Kerala floods of 2018** were among the worst floods in the state in nearly a century.

Key statistics:

- **30% excess rainfall during monsoon**
- **483 deaths**
- **5.4 million people affected**
- **1.4 million people in relief camps**
- **14,000 houses destroyed**

Economic losses exceeded **₹40,000 crores**.

A major contributing factor was the **simultaneous release of water from multiple dams** during intense rainfall.

The disaster demonstrated the growing impact of **climate change and extreme rainfall events**.



Community Role in Disaster Response — Kerala Example

One of the most remarkable aspects of the Kerala floods was the role of **local communities in rescue operations**.

Kerala fishermen played a crucial role in saving lives:

- Used their **fishing boats for rescue operations**
- Rescued **over 60,000 stranded people**
- Reached flood-hit areas before official rescue teams.

Technology also played a major role:

- **KeralaRescue.in platform**
- Google Maps for locating stranded people
- Social media coordination

This disaster demonstrated that **community participation is essential in disaster response**.



Need for Disaster Management Legislation in India

Before 2005, India did not have a comprehensive legal framework for disaster management. Disaster response was mainly **relief-oriented and reactive**.

Key characteristics of the pre-2005 system:

- No dedicated disaster management authority
- Disaster response handled under the **Famine Relief Code (1880)**
- Limited coordination between central and state agencies
- Focus mainly on **post-disaster relief rather than prevention**

Major disasters such as:

- **Odisha Super Cyclone (1999)**
- **Bhuj Earthquake (2001)**

highlighted the urgent need for a **national disaster management framework**.

This led to the enactment of the **Disaster Management Act, 2005**, which transformed India's disaster management approach.



Evolution to DM Act 2005



1999
Odisha Cyclone



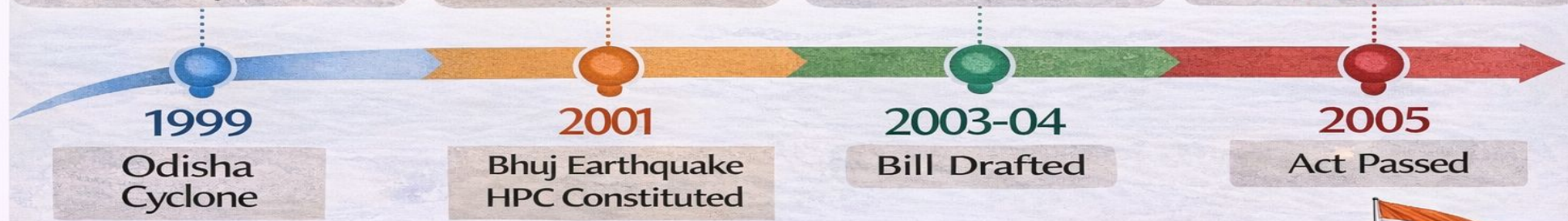
2001 Bhuj Earthquake
HPC Constituted



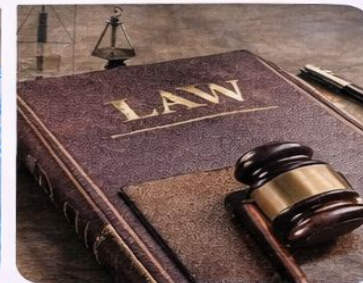
2003-04
Bill Drafted



2005
Act Passed



“From disaster to legislation” journey



Disaster Management Act, 2005

The **Disaster Management Act, 2005** is the primary legislation governing disaster management in India.

Important details:

- Passed on **23 December 2005**
- Enacted on **26 December 2005**
- Implemented after lessons from the **2004 Indian Ocean Tsunami**

The Act introduced a **paradigm shift** in disaster management:

Old Approach

Relief → Rehabilitation

New Approach

Prevention → Mitigation → Preparedness → Response → Recovery

Key objectives of the Act:

- Establish institutional structures for disaster management
- Improve coordination between central, state, and local authorities
- Promote disaster preparedness and risk reduction

Strengthen response and recovery systems.



Three-Tier Institutional Structure of Disaster Management

The Disaster Management Act established a **three-tier institutional structure** in India.

National Level

- National Disaster Management Authority (NDMA)
- National Executive Committee (NEC)
- National Disaster Response Force (NDRF)

State Level

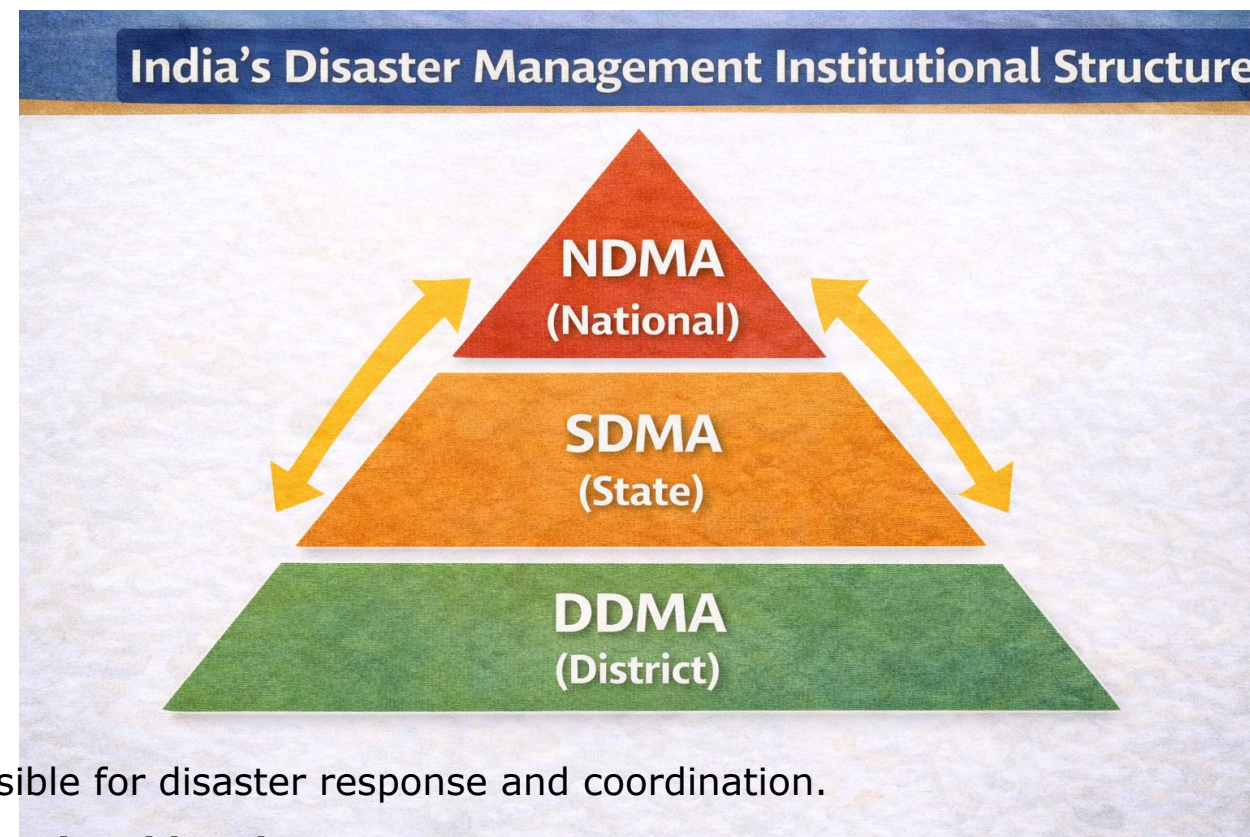
- State Disaster Management Authority (SDMA)
- State Executive Committee (SEC)
- State Disaster Response Force (SDRF)

District Level

- District Disaster Management Authority (DDMA)
- District Emergency Operations Centre (DEOC)

The **district administration is the operational unit** responsible for disaster response and coordination.

This structure ensures **vertical coordination from national to local level**.



National Disaster Management Authority (NDMA)

The **National Disaster Management Authority (NDMA)** is the apex body responsible for disaster management in India.

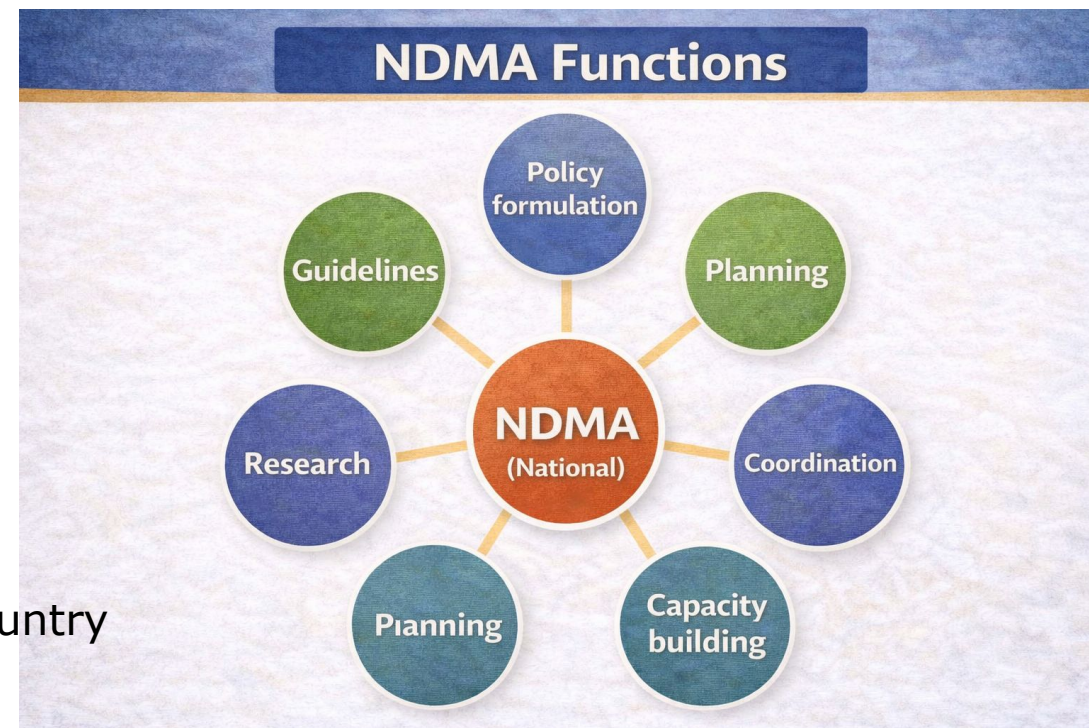
Key features:

- **Chairperson:** Prime Minister of India
- Maximum **9 members**

Major functions:

- Formulate national disaster management policies
- Approve the **National Disaster Management Plan**
- Issue guidelines to ministries and state governments
- Coordinate disaster management activities across the country
- Promote research, capacity building, and awareness.

NDMA plays a crucial role in developing **strategies for disaster risk reduction and preparedness.**



National Disaster Response Force (NDRF)

The **National Disaster Response Force (NDRF)** is India's specialized force for disaster response.

Key characteristics:

- Established in **2006**
- Comprises **12 battalions**
- Personnel drawn from paramilitary forces such as:
 - BSF; CRPF; CISF; ITBP; SSB

Specialized training areas:

- Urban Search and Rescue (USAR)
- Flood rescue operations
- Chemical, Biological, Radiological, and Nuclear (CBRN) emergencies
- Rope rescue and collapsed structure rescue
- K9 search operations

The NDRF has responded to **over 500 disasters** since its formation.



Financial Mechanisms for Disaster Management

India has dedicated funds to support disaster management operations.

State Disaster Response Fund (SDRF)

- Primary fund for disaster response at the state level
- Funded by both **central and state governments**
- Used for relief operations and emergency assistance

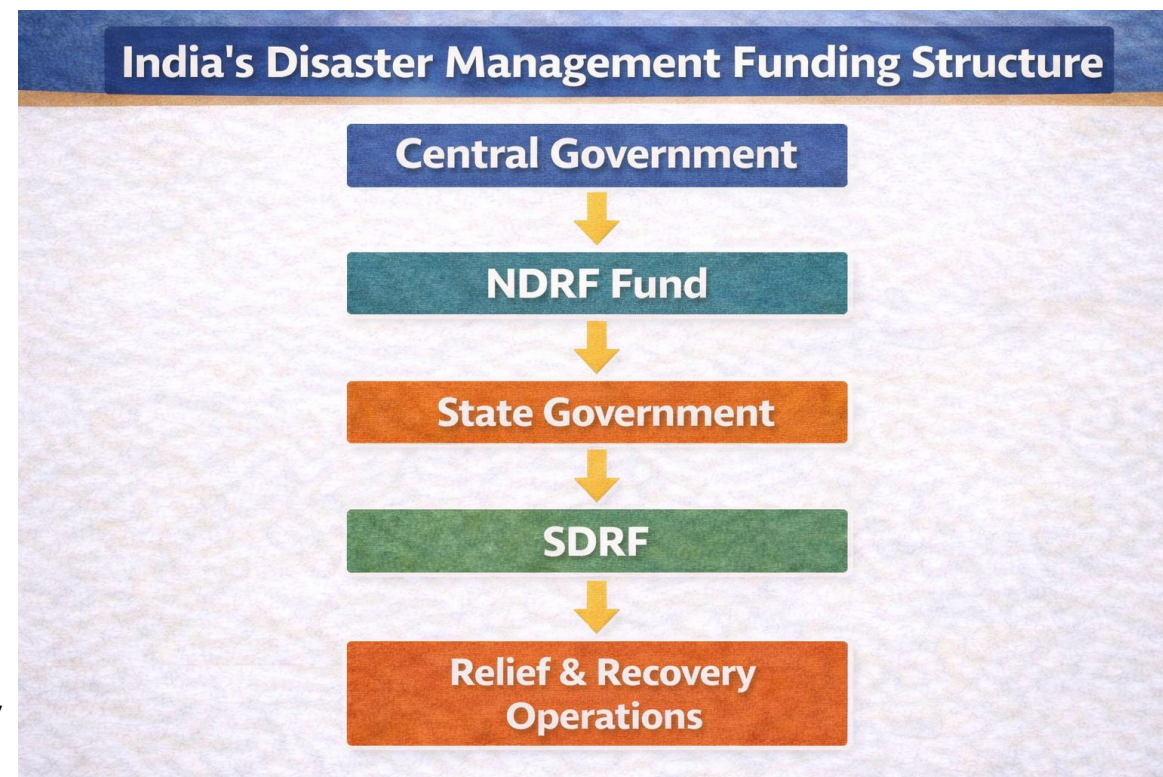
Examples of assistance:

- Compensation for deaths
- Housing support
- Relief camps and food distribution

National Disaster Response Fund (NDRF)

- Used for **severe disasters exceeding state capacity**
- Managed by the central government.

These funds ensure **financial support for relief and recovery activities.**



National Disaster Management Policy (2009)

The **National Disaster Management Policy (2009)** provides a strategic framework for disaster risk reduction.

Vision:

“Safe and disaster resilient India through proactive and technology-driven strategies.”

Key guiding principles:

- Community participation
- Multi-hazard approach
- Emphasis on prevention and mitigation
- Use of modern technology
- Coordination among government agencies
- Inclusion of vulnerable populations.



The policy promotes a **holistic approach to disaster management** involving all sectors of society.

National Disaster Management Plan (NDMP)

The **National Disaster Management Plan (NDMP)** is the operational plan for implementing disaster management strategies in India.

Key points:

- First released in **2016**
- Updated in **2019**
- Prepared by the **National Executive Committee**

The plan includes:

- Hazard risk assessment
- Preparedness strategies
- Response coordination
- Recovery and reconstruction plans

The NDMP aligns with the **Sendai Framework for Disaster Risk Reduction (2015–2030)**.

It serves as a **template for state and district disaster management plans**.

Sendai Framework for Disaster Risk Reduction



Role of Central Government in Disaster Management

The central government plays a crucial role in disaster management through:

Policy and Legislation

- Disaster Management Act (2005)
- National Disaster Management Policy
- National Disaster Management Plan

Early Warning Systems

Key national agencies include:

- IMD** – Weather forecasting and cyclone warnings
- CWC** – Flood forecasting
- INCOIS** – Tsunami early warning
- ISRO/NRSC** – Satellite monitoring and damage assessment

Financial Support

- Central government contributes to SDRF

Provides additional funds through NDRF during major disasters.



Role of State Governments

State governments are the **primary operational authorities** for disaster management.

Key responsibilities include:

- Preparation of **State Disaster Management Plan**
- Early warning dissemination
- Evacuation and rescue operations
- Management of relief camps
- Distribution of compensation and relief assistance
- Reconstruction and rehabilitation.

Examples of best practices:

- **Odisha:** Zero casualty cyclone mission
- **Gujarat:** Strong earthquake preparedness after Bhuj disaster
- **Kerala:** Improved flood preparedness after 2018 floods.

STATE DISASTER RESPONSE FLOW DIAGRAM



Role of Local Governments

Disaster management is most effective at the **local level**.

Local authorities such as **Urban Local Bodies (ULBs)** and **Panchayats** are responsible for:

- Maintaining drainage systems
- Enforcing building regulations
- Managing local evacuation and relief operations
- Preparing **local disaster management plans**

Village Disaster Management Committees (VDMCs) play an important role in:

- Hazard mapping
- Community awareness
- Disaster drills and preparedness activities.

Local participation ensures **quick response and effective risk reduction**.

Community Disaster Preparedness



Evacuation Route



Village Meeting



Community Volunteers

Role of NGOs in Disaster Management

Non-governmental organizations play a crucial role in disaster management due to their **community-level presence and flexibility**.

Key contributions include:

Preparedness

- Community training
- Awareness campaigns
- Risk mapping.

Emergency Response

- Search and rescue support
- Relief distribution
- Medical camps.

NGOs often act as **bridges between government and affected communities**.

Recovery and Rehabilitation

- Housing reconstruction
- Livelihood restoration
- Psychosocial counseling.

Examples of active NGOs:

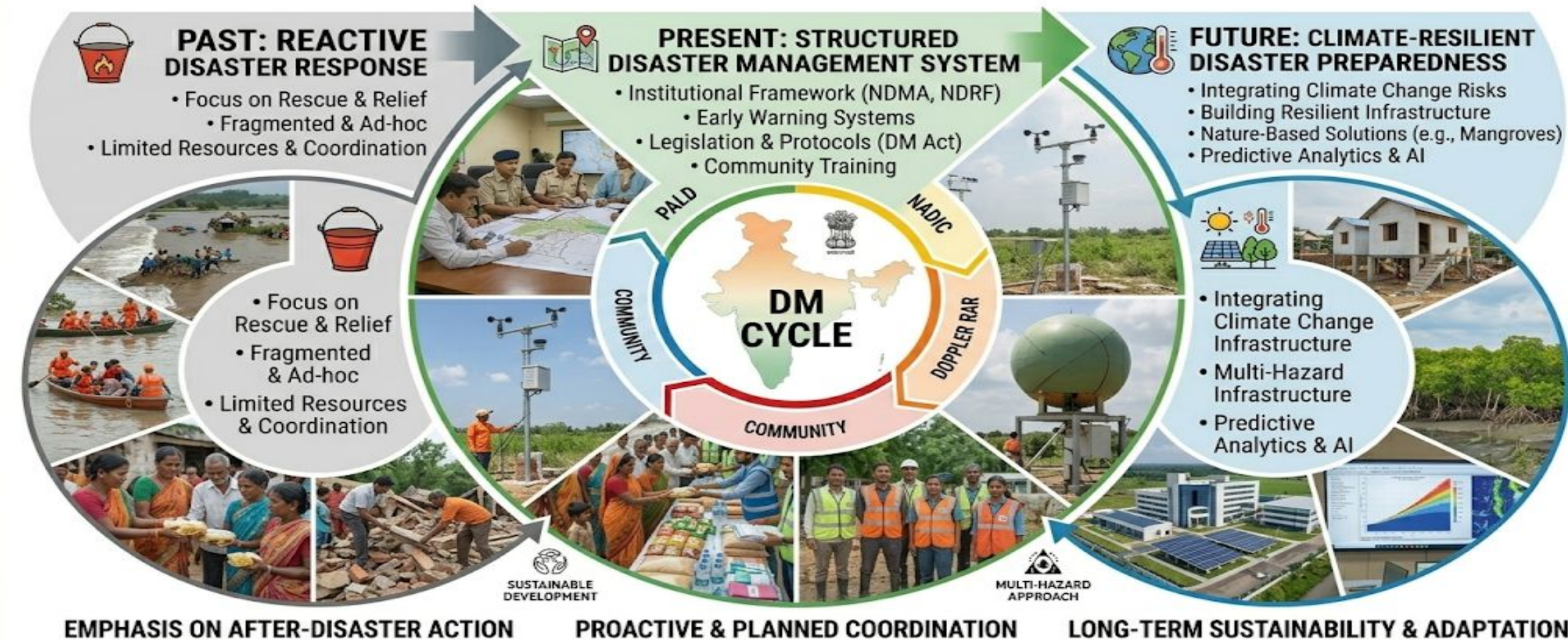
- SEEDS India
- Goonj
- Oxfam
- CARE
- ActionAid.

NGO DISASTER RELIEF DISTRIBUTION IN INDIA



India's Disaster Management Journey

DISASTER RISK REDUCTION PROGRESS TIMELINE IN INDIA





India's Disaster Management Journey

India's disaster management system has evolved significantly over time.

Pre-2005 Era

- Relief-based response
- Limited coordination
- No legal framework.

Post-2005 Era

- Disaster Management Act implemented
- NDMA and NDRF established
- Focus on prevention and preparedness.

Current Strengths

- Strong cyclone warning systems
- Specialized disaster response forces
- Improved technology and satellite monitoring
- Increasing community participation.

Despite progress, challenges remain in:

- Urban disaster preparedness
- Building code enforcement
- Climate change adaptation.



Connecting to Real-World Scenario

...to be continued if required.

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Interactive Activities

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Think-Pair-Share

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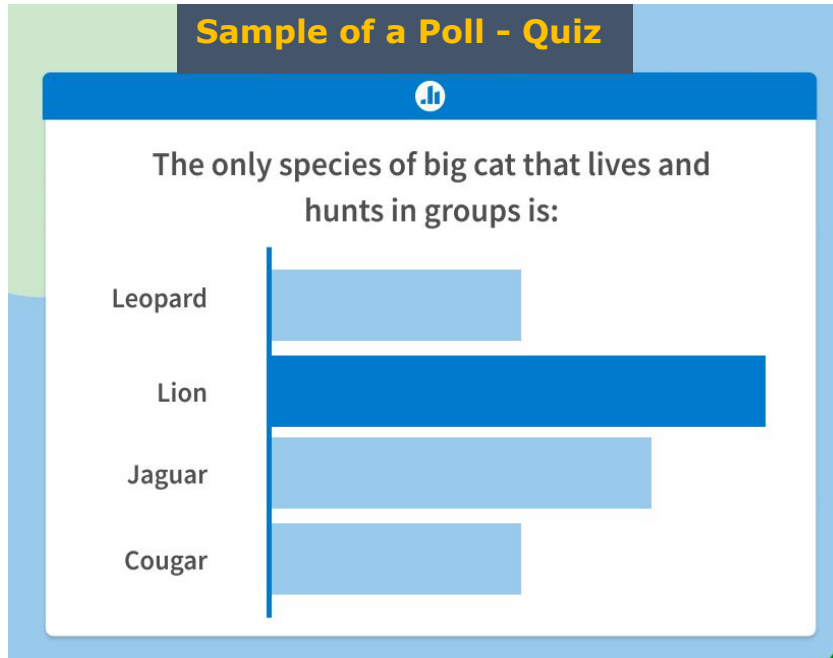
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Details of Interactive Activity



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GSFCU & Values

In every lecture, Faculty may talk on any of the following:

- **DCS & GSFCU Portal:** Remind students to use DCS, talk about it's features and instruct all to visit GSFCU website
- **Discipline, Values and Ethics.**
- **GSFC University – My University:** Emphasize the importance of taking ownership of the University.
- **Respect & Courtesy:** Remind students to be respectful to peers, teachers, and staff. Encourage polite language and behavior.
- **Punctuality:** Emphasize the importance of arriving on time to class and completing assignments by deadlines.
- **Classroom Rules:** Reinforce basic classroom rules, such as keeping mobile phones on silent and maintaining a clean environment.
- **Safety:** Remind students about safety protocols, whether it's physical safety in the classroom or digital safety when using online resources.
- **Positive Attitude:** Encourage a positive mindset, cooperation and supporting one another.

Any other thoughts related to Discipline, Values, Ethics etc. can be mentioned during these 2 minutes. You May use appropriate image / video with proper reference. One example is given in next slide.

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GSFCU & Values

Respect & Courtesy

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Dress
Appropriately



Respect
Personal Space



Be Mindful
of Your Words



Respect
Coworkers' Time



Keep Your
Workplace Clean



Socialise with
Your Peers



Use Your
Mobile Wisely



Take
Responsibility

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